Directional Structural Witnesses for Fail-Closed Cross-Domain Transfer

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Abstract

RAKL v3 can persist task episodes, consolidate scoped lessons and reuse experience without changing the underlying model weights, but the existence of a shared experience substrate does not determine when cross-domain reuse is valid. We study the transfer relation itself. A directional structural witness records context, quantity of interest, role mappings, preserved invariants, non-preserved properties, target boundaries, evidence and uncertainty, and is intended to license or reject movement of an episode, lesson or operator into a new problem fibre. In a deterministic conformance suite, the witness gate makes the intended decision on all six constructed Q2/Q3 cases across queue stability, positive feedback and threshold cascades; a deliberately semantic-only rule fails all six. We then froze and ran a leave-one-family-out diagnostic on 44 internally proposed Q1–Q4 pairs across 11 families. Witness features improved ROC-AUC by 0.086 and average precision by 0.097 over the strongest frozen lexical/tag controls, with Q2 true accept 1.00 and Q3 false accept 0.00. These are constructed, same-session proposal labels and do not establish an advantage over a modern content encoder. A fresh 16-item natural-domain source set and an opaque external-annotation packet are frozen, but zero external judgements or adjudications have been received; the exact strong semantic model asset is also not staged. The paper therefore treats the witness as a reproducible, fail-closed transfer formalism and a candidate evidential basis for v3 `TRANSFER_NOVEL` classifications, not as proof of natural cross-domain generalization, training-data efficiency or a universally valid learned skill substrate.

1 Problem: repeated payment for reusable structure

Language-model learning and inference both contain repeated work. During training, semantically different examples can instantiate the same dependency, conservation law, feedback motif, constraint pattern or reasoning routine. During inference, an agent can repeatedly reconstruct a procedure that was already solved in earlier trajectories. The tempting conclusion is that a system should externalize the repeated structure and reuse it. That conclusion is not novel. The research problem is to determine *what counts as the same reusable structure, when transfer is valid, and when the cost of inducing and verifying the structure is actually lower than relearning or rediscovering it.*

Several parent methods already occupy large parts of this space. Skill-It learns skill dependencies and uses them to adapt data sampling; in continual pretraining it reports a 3B-parameter model reaching higher evaluation-harness accuracy with 1B tokens than a uniform data-source baseline trained with 3B tokens [1]. MASS constructs a mathematical skill graph for pretraining-data selection and reports comparable downstream mathematical performance after removing substantial fractions of the training tokens [2]. These results make "graph-guided data selection saves training data" an inherited mechanism, not a RAKL contribution.

The inference side is similarly mature. Reasoning Primitive Induction mines successful ReAct traces, clusters recurrent moves and compiles them into typed pseudo-tools that improve several held-out reasoning and planning tasks at lower average inference cost [4]. TraceCompiler mines noisy tool-use traces into mostly deterministic workflows with evidence-bearing producer–consumer dependencies and can refuse compilation when an irreversible branch is underdetermined; importantly, it reports call reduction without claiming net efficiency because offline compilation cost was not measured [5]. SWIFT directly frames agent-workflow design as amortized transfer from structural priors and reports large reductions in marginal per-task workflow-search cost across source and unseen tasks [3]. Its operator-name randomization result further suggests that topological structure, not just semantic labels, carries much of the transfer signal.

Abstraction itself is also an active target. Controlled ACL experiments study whether language models learn structural information and can compose it at test time [6]; RLAD trains an abstraction generator and abstraction-conditioned solver rather than relying only on long brute-force traces [14]; related abstraction-reinforcement work reports improved robustness and out-of-distribution reasoning [15]. The candidate contribution here is therefore not "LLMs need abstraction."

1.1 Residual research question

The residual is a conjunction not established by the parents above. We ask whether one persistent structural object can simultaneously support:

1. cross-domain training-data redundancy judgments when surface semantics differ;
2. test-time retrieval or adaptation of a previously learned operator;
3. explicit rejection of semantically similar but structurally invalid analogies;
4. boundary- and QoI-scoped transfer rather than global similarity;
5. evidence-bearing failure history so negative transfer changes future reuse; and
6. total cost accounting that includes the one-time induction and verification cost.

This makes the paper mechanism-seeking rather than leaderboard-seeking. If a skill graph, workflow prior or semantic retriever already supplies the same predictive transfer signal at lower cost, the RAKL structural layer has not earned its complexity. Conversely, a positive result is strongest when the same structural identifier is learned or stored once and measurably improves both data selection and later inference on domains whose surface vocabulary would not have linked them directly.

1.2 Shared experience banks and dependency-aware skill retrieval narrow the residual further

A closer parent for the "learn once, reuse across tasks" intuition is ReX, which treats recurring reasoning patterns and skills as latent experiences rather than task-specific adapters [7]. ReX learns a shared Experience Bank of foundational skill vectors, encodes each input into a low-dimensional experience code, and dynamically composes the relevant skill vectors into a lightweight adapter without requiring explicit task identifiers. Its multi-task results show that cross-input/task experience reuse can improve specialization and generalization. This removes any novelty claim that Paper 3 uniquely proposes a persistent shared experience substrate or dynamic composition of reusable skills.

Dependency-aware skill retrieval is also moving beyond flat semantic matching. SkillGraph represents agent skills as a dependency graph and targets selection/composition failures that arise when modular skills are treated as isolated text descriptions [8]. SKILLGRAPH goes further by representing reusable skills as an evolving typed graph whose prerequisite, enhancement and co-occurrence edges are updated from trajectories and reinforcement-learning feedback; ordered skill subgraphs guide new-task execution while the same graph participates in improving the policy [9]. GraSP compiles retrieved skills into executable typed DAGs with precondition–effect edges, node-level verification and locality-bounded repair [10]. AgentGL similarly treats graph topology as an active reasoning substrate: an LLM agent uses graph-native tools at inference while a graph-conditioned curriculum is used during reinforcement learning [12]. These systems are important parent mechanisms because they show that relational topology can inform learning, retrieval, composition and execution.

A complementary retrieval parent exposes why the negative controls must be strong. SkillSight shows that recurring descriptive boilerplate across skill documents can inflate dense and lexical relevance scores even when those shared words are not discriminative of the required capability; calibrating that background improves retrieval without extra model training [11]. Paper 3 therefore cannot treat a semantic retriever failure as evidence for structural novelty unless the semantic baseline is already calibrated against shared-background bias.

The candidate residual must therefore remain more specific. Paper 3 asks whether a *scientifically scoped* structural object—with explicit QoI, context, directional role mapping, preserved and non-preserved properties, boundary conditions, evidence and negative-transfer history—can do two jobs with the same persistent identity: estimate *cross-domain training-data redundancy* and license or reject inference-time transfer. ReX supplies reusable latent experience; SkillGraph/SKILLGRAPH/GraSP supply increasingly rich dependency and execution graphs; AgentGL demonstrates one graph-conditioned learning/execution loop; SkillSight supplies a stronger semantic-retrieval control. RAKL must either add value in the controlled surface-disjoint, boundary-sensitive training-selection plus inference regime or narrow its claim to a scientific-validity governance layer rather than a general reuse advance.

Post-v3 residual. RAKL v3 now implements the persistent episode/lesson/tool substrate that earlier drafts treated as part of the broader motivation. That implementation narrows the paper rather than strengthening its empirical claim. The remaining question is whether the directional witness is a useful *transfer-validity relation* inside such a substrate. Persistence, lesson consolidation and shared retrieval are therefore architectural premises supplied by the framework; boundary-sensitive cross-domain licensing remains the scientific object tested here.

2 A scoped structural transfer object

A reusable structure should not be a bag of similar words. Let a structural object be

$$S = (V, E, \chi, \gamma, q, \varepsilon),$$

where V is a set of typed roles, E a set of typed directed or undirected relations, χ a set of invariants or constraints, γ a boundary/context record, q the quantity of interest, and ε the evidence identities supporting the extraction. Domain-facing entities are retained in provenance, but the transfer layer reasons over roles and relations.

The object is intentionally scoped. A hospital and a packet network need not be globally "the same." They may instantiate the same queue-stability substructure for a backlog or delay QoI while differing radically in ethics, priority policies, intervention constraints and entity semantics. The formalism therefore uses a directional mapping witness rather than a global equivalence relation.

Definition 1 (Directional structural witness)**.** *For source S_A and target S_B, a witness is*

$$w_{A \to B} = (\mu_V, I^+, I^-, B, E_w, U_w),$$

where μ_V maps source roles to target roles, I^+ records preserved invariants, I^- records explicitly non-preserved properties, B contains required target boundary conditions, E_w is evidence for the mapping and U_w records uncertainty. The witness is licensed only for the registered QoI.

The relation is directional because a transformation useful in A may require information or interventions unavailable in B. It is context-specific because a mapping that is valid in one regime can fail after a boundary change. It need not be transitive: $A \to B$ and $B \to C$ do not automatically license $A \to C$ unless the composed invariant and boundary obligations are separately satisfied.

2.1 Transfer gate

The reference transfer-gate module checks four non-compensatory conditions. First, the source and target endpoints must match the witnessed objects. Second, every required source relation must survive under the role mapping in the target. Third, the source invariants required for the intended operation must be declared and present in the target. Fourth, every required target boundary must match. QoI mismatch is itself a rejection. Domain labels and semantic similarity are not positive license conditions.

This gate is conservative by design. It can return `CANNOT_CHECK` for an invalid witness identity and `REJECTED` for an examined but failed mapping. A low semantic similarity is not a reason to reject when the relations and boundaries are witnessed; a high semantic similarity is not a reason to accept when the load-bearing invariant fails.

2.2 Why explicit non-preservation matters

Analogies are dangerous when the reusable part and the non-reusable part are stored in one undifferentiated similarity score. The witness therefore records I^-. In the queue example, "arrival competes with service and excess load grows backlog" may transfer while domain-specific priority policy and entity semantics do not. This is more than documentation: a later operator can declare which preserved invariants it depends on, and an intervention touching a non-preserved property can fail closed.

2.3 Operators as executable compressed knowledge

A later RAKL layer can associate structural classes with operators

$$T = (\text{Pre}, \text{Transform}, \text{Post}, B, C, F, E_T, A_T),$$

for preconditions, executable transformation, postconditions, boundaries, cost, failure modes, evidence and allowed authority effects. Paper 3 does not yet claim useful autonomous operator invention. The present objective is more basic: determine whether the structural object can safely retrieve or adapt an already validated operator across surface-disjoint domains. Operator induction becomes a later extension only after this transfer mechanism survives the benchmark.

2.4 Empirical evidence argues against symmetric structural equivalence

The directional witness is not only a conservative formal choice. Wang reports a controlled asymmetry in structural transfer between natural-language and biological foundation models [13]. Language models trained or fine-tuned on structural language tasks retain meaningful competence when moved to biological-sequence structure, while the reverse transfer remains weak across matched-token controls, scaling and multiple biological model families. The result is domain-specific and does not establish a universal law of transfer, but it is a direct counterexample to the assumption that shared structural regularity implies symmetric representational transfer.

For Paper 3 this has two consequences. First, a structural relation should default to a directed edge $S_A \to S_B$ with a witness rather than an undirected equivalence class. Second, transfer evaluation should include direction reversal as an explicit perturbation. If $A \to B$ succeeds but $B \to A$ fails, the library should preserve that asymmetry rather than canonicalize the two tasks into one globally interchangeable class. Any later transitive composition likewise requires a new scope/invariant check.

This also strengthens the Q2/Q3 benchmark logic. The object being tested is not whether two domains "look analogous" in a human narrative sense. It is whether a particular source-side operator or inference depends only on relations and invariants that are actually available in the target under the registered boundary conditions. Directional negative-transfer history is therefore part of the reusable state.

3 Directional witnesses as RAKL v3 experience-transfer relations

RAKL v3 makes experience persistence an implemented architectural fact but leaves transfer validity as an evidence question. A source `TaskEpisode` says what happened in one registered context. A versioned `Lesson` is a scoped abstraction over one or more episodes. Neither object is automatically applicable to a new target. The role of the directional structural witness is to supply an explicit candidate relation

$$W^{(q)}_{s \to t} : (E_s, L_s, \gamma_s) \rightsquigarrow (P_t, q, \gamma_t)$$

whose obligations state which roles and invariants are preserved, which properties are not preserved, and which target boundaries would invalidate reuse.

This makes the witness directional in two senses. First, the target quantity of interest determines which source structure is load-bearing. Second, a valid mapping from source to target need not license the reverse mapping because scope, information loss or boundary conditions can be asymmetric. A witness is therefore not an embedding similarity score and not an equivalence class over tasks.

3.1 Transfer can alter search priority without creating authority

When a witness passes the registered structural and boundary gate, v3 may use the linked episode or lesson to increase retrieval priority, activate an already promoted tool, or place an operator path earlier in the search queue. The permitted inference is

$$\text{witnessed applicability} \Rightarrow \text{eligible experience reuse,}$$

not

$$\text{witnessed applicability} \Rightarrow \text{scientific claim true.}$$

Scientific authority remains owned by the target problem's verification path. This is the cross-domain specialization of the framework invariant that experience-conditioned routing is not epistemic authority.

3.2 Negative transfer is learned through an evidence ladder

A failed transfer attempt first creates a non-success task episode. It does not immediately establish that the witness class, operator or source lesson is invalid. The system records the failure as observed-only, then permits competing diagnoses such as context mismatch, broken invariant, implementation defect, missing evidence or an unrelated downstream failure. Only discriminating evidence can promote a supported diagnosis, and only fresh replay/transfer can turn that diagnosis into a reusable boundary lesson.

The resulting discipline prevents one near-miss from becoming a global blacklist while still allowing repeated boundary-sensitive failures to improve later routing. It also makes negative-transfer history addressable: a future witness can explicitly inherit a known non-preserved property or target boundary rather than merely receiving a lower opaque score.

3.3 Connection to RAKL-triviality

V3 classifies a verified solution as `TRANSFER_NOVEL` when no new primitive was invented but pre-existing problem-solving structure was transported into a new context by an explicit mapping witness. The directional witness studied here is a candidate evidence object for that classification. A solution should receive `TRANSFER_NOVEL` only when the transfer relation is explicit, the target solution is independently verified, and resource ancestry shows that no new representation/operator/ontology was required.

This gives Paper III a narrower post-v3 claim. The framework already possesses a persistent experience substrate; this paper does not claim novelty for persistence itself. It asks whether one scientifically scoped, boundary-aware relation can make cross-domain reuse safer and more measurable than semantic resemblance alone. The natural-domain, strong-control and independent-annotation gates remain necessary before that question can receive an empirical answer.

4 A semantic–structural factorial benchmark

The central benchmark independently manipulates surface-semantic similarity and structural similarity. Let $S_{\text{sem}} \in \{\text{low}, \text{high}\}$ and $S_{\text{str}} \in \{\text{low}, \text{high}\}$. The four cells are:

Cell	Semantic relation	Structural relation
Q1	high	high
Q2	low	high
Q3	high	low
Q4	low	low

Q2 and Q3 are the load-bearing cells. Q2 asks whether the system can transfer when vocabulary and domain semantics are deliberately distant but the relevant relation/invariant structure is preserved. Q3 is a semantic decoy: vocabulary is similar and embedding similarity should be high, but the causal/dynamical structure or boundary condition that justifies transfer is changed. A useful structural system should accept Q2 and reject Q3.

The deterministic scaffold in `src/rakl/structural_benchmark.py` now exercises this requirement across three distinct mechanism families rather than one illustrative analogy. For *queue stability*, a computer network and an emergency department use different entity vocabularies but share a continual-arrival/finite-service/backlog structure for the registered QoI; the high-semantic decoy instead uses queue language in a finite-batch regime, so the steady-flow stability invariant is invalid. For *positive feedback*, viral adoption and a bank-run withdrawal loop share the sign structure required for perturbation amplification, while a semantically similar control-feedback target reverses one loop edge and becomes stabilizing negative feedback. For a *threshold cascade*, epidemic spread and cascading grid outage share a supercritical propagation structure, while a same-domain subcritical cascade decoy removes the load-bearing reproduction relation and violates the threshold/connectivity boundary. The original Q4 case remains a semantically and structurally unrelated queue-to-feedback mapping.

4.1 Deterministic conformance result

The exact-subject executable receipt contains six load-bearing Q2/Q3 cases: one Q2 and one Q3 case in each of the three mechanism families. The deliberately semantic-only decision rule licenses high-semantic candidates and rejects low-semantic candidates; it is therefore wrong on all six hard cases by construction. The structural gate instead checks the registered directional witness, QoI, load-bearing relations, invariants and target boundaries. It licenses all three Q2 transfers and rejects all three Q3 decoys.

Decision rule	Q2 valid licensed	Q3 invalid accepted	Hard cases correct
Semantic-only control	0/3	3/3	0/6
Structural witness gate	3/3	0/3	6/6

This table is a software-conformance result, not a statistical generalization result. The cases were deliberately constructed to separate semantic and structural similarity; their role is to establish that the implementation can express the intended distinction before model-level experiments are authorized. They provide no estimate of performance on naturally occurring cross-domain mappings.

These cases are deterministic conformance worlds, not an empirical claim that the representation generalizes. Their purpose is to prevent a one-family implementation from earning an expensive training run merely by memorizing queue-specific details. The confirmatory dataset will still require independent annotations and broader families such as flow conservation, bottlenecks, delayed negative feedback, diffusion/contagion, matching, scheduling, concentration fragility, path dependence and search/backtracking. Each family should span multiple surface domains and include adversarial near-misses.

4.2 A broader internal diagnostic fails closed at the annotation gate

Before inspecting predictor outcomes, we froze a balanced Q1–Q4 proposal set, leave-one-family-out split, predictor family, metrics and stopping thresholds at Git subject `f2701f732f83`. The resulting diagnostic contains 44 constructed proposal pairs across 11 proposed mechanism families. Every coordinate was proposed in the same assistant session that assembled the benchmark. None has two independent human/expert annotations or adjudication, so all 44 are explicitly marked ineligible for confirmatory use.

The local pilot fits deterministic L2-regularized logistic predictors using successively richer feature sets: surface overlap, skill-tag overlap, dependency-pattern overlap and witnessed structure. It is a cheap predictive diagnostic with no foundation-model judgement. Leave-one-family-out outcomes crossed the frozen incremental-signal thresholds: relative to the strongest of these frozen lexical/tag controls, the dependency-aware model, witnessed structure increased ROC-AUC from 0.914 to 1.000 and average precision from 0.903 to 1.000, while reducing Brier loss from 0.166 to 0.039. Its Q2 true-accept rate was 1.00 and Q3 false-accept rate was 0.00. These point estimates measure discrimination of labels encoded in an internally constructed proposal set; they do not estimate human agreement, natural transfer performance or incremental value beyond a modern content encoder.

The independent-annotation gate nevertheless has authority over the diagnostic-signal gate. Its exact verdict is `FAIL_CLOSED_MISSING_INDEPENDENT_ANNOTATION`: zero of 44 items are confirmatory-eligible. Consequently, no training or inference run was launched. The unresolved residual is evidence acquisition, not a license to substitute same-session agreement or spend compute defending the original shared-substrate headline.

A subsequent chronology audit identified a stricter cut: the frozen v1 protocol records both that labels were visible during construction and that confirmatory use is not permitted. The 44-item v1 set is therefore permanently an internal constructed diagnostic; later annotations

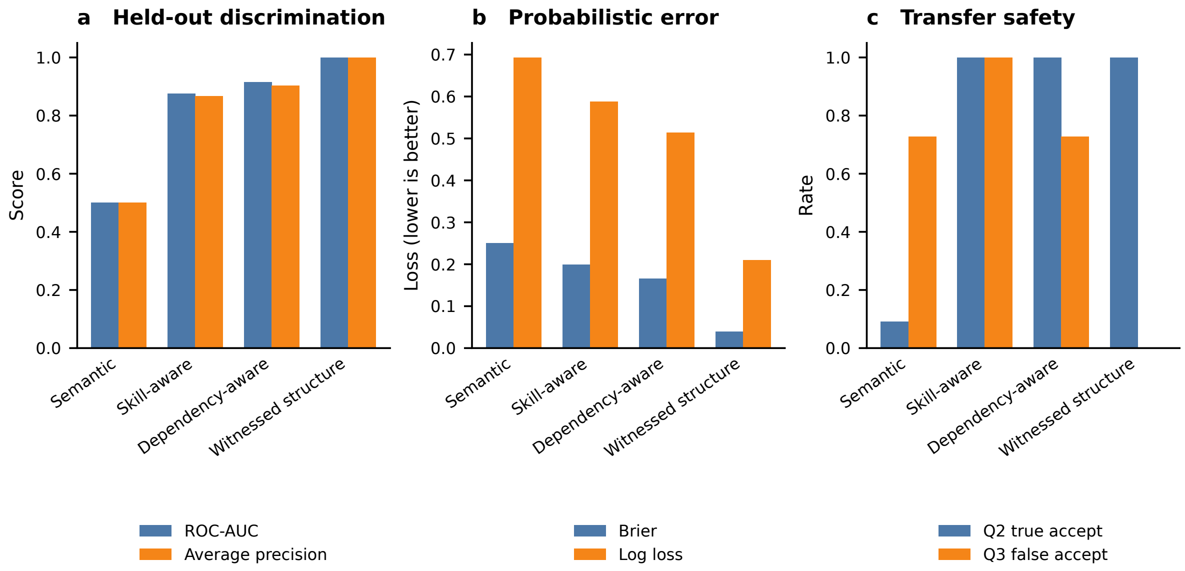

Figure 1: Receipt-bound internal cheap-gate diagnostic. The unit is a constructed source–target pair ($n = 44$) and each prediction is out-of-family under a deterministic leave-one-family-out fit. Bars are aggregate point estimates; there are no repeated seeds, sampling intervals or independent annotations. Panel a reports ROC-AUC and average precision, panel b Brier and log loss, and panel c Q2 true accept with Q3 false accept. The figure is generated only from the checked machine-readable result receipt; values are not hand-entered into the plot.

cannot retroactively convert it into confirmatory evidence or authorize compute. We retain that receipt unchanged as negative history and register a separate v2 lane. V2 requires a fresh label-blind item set frozen before labels or results, the same or stricter signal thresholds, two genuinely independent human/expert annotations per item, distinct post-freeze adjudication and an external provenance audit. Repository validation can check hashes, coverage, chronology and attestations, but cannot itself prove that an independence attestation is truthful. Until both the new annotation gate and held-out signal gate pass, the LUNARC preflight refuses training and inference submission.

The first v2 curation tranche now freezes 16 fresh label-blind source–target descriptions across four previously unused families: bottleneck–minimum-cut, diffusive transport, matching and allocation, and path dependence and reinforcement. The artifact binds 19 citation records and preserves four items per family, but it contains zero external judgements, no adjudication and no signal estimate. Its role is chronological: it fixes what future annotators will assess before their answers exist. Consequently no training or inference job is authorized; the writable FS9 workspace and GPU-account observation are execution preparation only.

A pre-annotation identifier-leakage audit then found that four curator-only source identifiers included the phrase `near_miss`. Although those identifiers were not outcome fields, they could reveal a structural expectation if an annotator inspected the source artifact or reconstructed the linkage. We therefore preserve that first source set as negative history and replace it with an otherwise byte-equivalent v2.1 set whose identifiers are neutral sequential codes. The earlier packet attempt was discarded and its private linkage removed. This repair was completed with zero external judgements and was superseded before any external judgement; no training or inference job is authorized.

4.3 Structural score must add information beyond semantic score

For a transfer outcome Y, the confirmatory analysis compares models such as

$$\operatorname{logit} P(Y = 1) = \beta_0 + \beta_1 S_{\text{sem}}$$

against

$$\operatorname{logit} P(Y = 1) = \beta_0 + \beta_1 S_{\text{sem}} + \beta_2 S_{\text{str}} + \beta_3 S_{\text{sem}} S_{\text{str}} + u_{\text{family}}.$$

The central mechanism requires material incremental predictive value from the structural term and, operationally, a low invalid-transfer rate in Q3. If a strong semantic retriever or LLM analogy judge already predicts valid transfer as well as the witnessed structure, the proposed representation does not earn an additional layer.

The frozen confirmatory item set holds 16 items, and at that size a bare point-estimate threshold cannot distinguish signal from sampling noise. The Hanley–McNeil paired standard error for an AUC difference over 16 items ranges from 0.067 to 0.176 across plausible inter-arm correlation and label-balance assumptions, so a frozen gain threshold near 0.05 sits within one third of a standard error of pure noise and returns the wrong verdict roughly one time in three, in both the false-pass and false-fail directions. The Q2 and Q3 acceptance rates are proportions over approximately eight items each, where one item moves either rate by 0.125 and the 95% interval half-width at $p = 0.5$ is ± 0.35 against thresholds of 0.8 and 0.2. A point estimate alone is therefore uninterpretable at this sample size.

The confirmatory gate consequently carries an uncertainty-quantification layer that the protocol froze before any annotation arrived. For each item the per-item Brier score is computed for the witnessed-structure arm and the strongest control arm under the shared adjudicated label, and the paired difference (control minus structural) forms the resampling unit. A paired bootstrap confidence interval and a sign-flip permutation p-value, both implemented in a standard-library inference helper, are evaluated with a frozen seed and 10,000 resamples. The diagnostic gate passes only when this interval excludes zero in addition to every frozen point-estimate threshold; a point estimate that clears 0.05 while its interval overlaps zero is reported as indistinguishable from noise rather than as confirmation. This is an evaluation-design repair: it changes what the gate can say about a future annotated outcome, and no result is claimed, supported or refuted here.

4.4 Annotation and leakage

A human-curated version will record role graphs, typed relations, invariants, boundary conditions and QoI with at least two independent annotations plus adjudication. Inter-annotator agreement is reported separately for roles, relations, class assignment and boundaries. The final domain split must prevent a surface domain from appearing in both training and target evaluation for the same structure family. Synthetic generation can expand coverage, but held-out human adjudication is required to show that the structural labels are not merely artifacts of the generator.

4.5 Fresh v2.1 annotation packet

After the neutral-identifier repair merged, we generated a new packet from the exact v2.1 source set and bound it to merged Git subject `f4cee8313ec64d02873b87f92c51c35c113cd70d`. The public packet contains 16 opaque item identifiers and no curator source identifiers, outcomes, annotations, adjudication or signal result. A coordinator-only linkage is held outside Git on FS9 with mode `0600`; the public receipt records only its location and content hashes. The discarded pre-repair packet was not reused.

A same-context hostile usability check found that the original response schemas forced `cannot_assess=false`, despite the frozen rubric instructing annotators not to guess under missing evidence. The repaired schemas permit `cannot_assess=true` only when all nine judgement coordinates are null. The compiler preserves that submission but fails confirmatory import, so missing evidence remains visible negative history rather than a coerced label. No external judgement exists at this packet-freeze stage, and training and inference remain unauthorized.

4.6 A content-bound semantic control is frozen before labels

The v2 evaluator audit exposed a weak-control residual: surface Jaccard, skill-tag Jaccard and dependency-tag Jaccard are transparent conformance controls, not a strongest-feasible modern semantic comparator. Before any external judgement became visible, we therefore froze a separate content-bound control using the exact revision `953dc6f6f85a1b2d` of `BAAI/bge-reranker-v2-m3`. Its input is a deterministic source/target rendering of the v2.1 domain, surface terms, skill tags, dependencies, citation/title strings and shared QoI. Every rendered side and pair receives a SHA-256 binding. Candidate invariant/boundary proposals, annotations, quadrants and transfer outcomes are excluded, so the control cannot silently consume the structural target it is meant to challenge.

This is a protocol freeze rather than a result. An earlier clean-environment descriptor attempt returned `CANNOT_CHECK_MODEL_ASSET_MISSING` because the 2.27-GB model artifact was absent from that checkout and emitted zero scores. Subsequent FS9 harvests (jobs 3476527–3476529; frozen descriptor-harvest status receipt under `research/`) reached `HARVEST_DESCRIPTOR_READY` with sixteen label-blind records while keeping `training_authorized=false`. Repository issue `#217` still records zero public annotator, adjudicator or provenance-auditor responses, so the zero-label chronology observation remains in force. Descriptor readiness remains execution preparation only: it does not mint labels, authorize training or inference, or support confirmatory Paper III metrics. The eventual signal test must still await external annotations and adjudication, and must compare witnessed structure against the strongest held-out non-structural arm including the reranker and its skill/dependency augmentation; selecting the older lexical control after seeing labels is forbidden.

5 Structural amortization as a cost-to-capability hypothesis

Reuse is not free. Let C_I be the one-time cost of inducing, validating and storing a reusable structure. Let c_B be the baseline per-task cost of solving without reuse and c_R the per-task cost of retrieving, adapting and verifying the reusable structure. Then

$$C_B(n) = nc_B, \qquad C_R(n) = C_I + nc_R.$$

When $c_R < c_B$, reuse becomes strictly cheaper once

$$n > \frac{C_I}{c_B - c_R}.$$

The executable reference in `src/rakl/amortization.py` reports this break-even count and returns no finite break-even when the reuse path is not cheaper per task. This elementary accounting is important because prior workflow-compilation work can reduce runtime calls without establishing net efficiency when offline compilation cost is omitted [5].

The empirical paper uses a fuller vector

$$C_{\text{total}} = C_{\text{induction}} + C_{\text{training}} + C_{\text{retrieval}} + C_{\text{adapt/reason}} + C_{\text{tools}} + C_{\text{verification}}.$$

No coordinate is silently set to zero. A cost-to-capability point is admissible only if it reaches the frozen capability target without crossing the registered validity-failure ceiling.

Hypothesis 1 (Structural transfer). *At matched semantic similarity, a witnessed structural score provides material incremental prediction of held-out transfer success.*

Hypothesis 2 (Semantic-decoy rejection). *A witness-aware transfer gate has a lower invalid-transfer rate than semantic retrieval or analogy-only baselines on Q3 cases with high semantic and low structural similarity.*

Hypothesis 3 (Training efficiency). *At a frozen structural-OOD capability target, structural curation reaches the target with fewer training tokens or examples than strong semantic/diversity and skill-graph parents in regimes containing cross-domain structural redundancy.*

Hypothesis 4 (Inference efficiency). *For a frozen base model and target capability, reuse of validated structural objects/operators reduces test-time reasoning/search cost relative to semantic memory, analogy prompting and strong skill/workflow-prior parents after all retrieval, adaptation and verification costs are included.*

Hypothesis 5 (Redundancy mechanism). *Under a controlled data generator, the advantage of structural reuse increases with structural redundancy after surface-semantic diversity is held approximately fixed.*

Hypothesis 6 (Shared substrate). *A structure identifier induced or registered during training/data curation can be reused at inference without changing to an unrelated task-specific representation. The strongest paper claim requires this shared object; two independent tricks would support two narrower results rather than one unification.*

Each hypothesis has a direct falsifier. Flat structural coefficients, high Q3 transfer, no break-even cost, a strong MASS/Skill-It/SWIFT-like parent matching RAKL, or incompatible training/inference representations all narrow the paper.

6 Evaluation programme

6.1 Training arm

The first phase uses small open models or controlled transformers so that the structural-redundancy mechanism can be tested before expensive scaling. Training datasets are generated or curated at controlled structural-redundancy levels while token length, difficulty and surface-semantic diversity are held as stable as feasible. Baselines include uniform sampling, exact/semantic deduplication or diversity selection, Skill-It-style dependency sampling, and MASS-style skill-graph selection. The RAKL arm differs only in using cross-domain structural objects and witnessed structural novelty rather than predefined surface skill labels.

Primary outcomes are tokens/examples to a frozen structural-OOD capability target, total training compute where measurable, performance on structure-known/domain-new tasks, performance on genuinely structure-new tasks, calibration and negative transfer. A positive result on ordinary in-domain accuracy without structural OOD does not support the central claim. Skill-It can carry parent authority only if its data-driven dependency learning and sequential adaptive sampler are reproduced under matched model, corpus, token budget, evaluation schedule and seeds; static dependency overlap is not a faithful substitute. MASS is a mathematical-pretraining parent and requires a parent-compatible mathematical skill graph, selection rule and matched mathematical evaluation. A cross-domain adaptation outside that scope is reported as an adaptation diagnostic, not direct reproduction.

6.2 Inference arm

The main inference experiment freezes the base model. Baselines include vanilla reasoning, semantic exemplar/memory retrieval, analogy prompting, a reusable reasoning-primitive library and, only where tasks expose a shared executable operator interface, a structural workflow prior in the style of SWIFT. The RAKL arm retrieves a structural object, validates a directional witness, retrieves or adapts a validated operator where one exists, and verifies the target result. Outcomes include capability, Q2 transfer, Q3 invalid-transfer rate, reasoning tokens, latency, search nodes, tool calls and verification cost. A faithful SWIFT comparison matches the base model, operator library, source workflows, workflow-search budget, interface, verifier and execution budget and charges prior-construction cost. For scientific pairs without an executable workflow interface, SWIFT is marked not applicable rather than weakened into a graph-similarity scalar.

6.3 Controlled structural redundancy

A synthetic or semi-synthetic generator produces datasets with increasing effective duplication of underlying structures while surface forms remain varied. The preregistered curve is

$$G(\rho_S) = \mathrm{CTC}_{\mathrm{parent}} - \mathrm{CTC}_{\mathrm{RAKL}},$$

where ρ_S is an uncertainty-aware structural-redundancy measure and CTC is cost to the frozen capability target. The analysis tests a predeclared monotone or rank relationship rather than choosing a curve after observing results.

6.4 Parent-method assimilation

Each strong baseline is first treated as a parent method. Its strongest relevant mechanism is reproduced as faithfully as feasible and mapped into the RAKL atlas. If the mechanism is useful, the final system may assimilate it; the scientific comparison then asks whether the residual RAKL structure adds value beyond the parent. For example, MASS and Skill-It become training-selection parents, SWIFT becomes a workflow-structure parent, and Reasoning Primitive Induction/TraceCompiler become procedural-reuse parents. The project does not intentionally weaken them to preserve novelty.

The cheap signal gate separately requires a modern semantic comparator. Its frozen control consumes the exact label-blind source and target descriptions with a hash-bound BGE cross-encoder while excluding proposed structural witnesses and all outcome fields. Missing or mismatched model files, content hashes, cases or pre-label chronology return `CANNOT_CHECK`; surface Jaccard cannot be substituted. No descriptor or result exists at this protocol-freeze stage because the exact model asset has not been staged.

Native execution is additionally frozen as two distinct LUNARC CPU batches. An allocated staging job must download the exact public revision and locally verify all six file sizes and SHA–256 values before atomic promotion. A separate descriptor batch is submittable only after the scheduler-bound stage harvest passes for the same exact clean merged checkout. It runs offline with local files only, requires the fast-tokenizer pair probe, and requires the shared runtime tree to equal the exact digest first attested by the allocated staging job and to remain unchanged through inference. A later descriptor harvest is authoritative only with a payload-free post-descriptor zero-label observation or a first-label cutoff proving that the descriptor timestamp precedes label availability. The freeze-time zero-label receipt is preflight evidence only. At this contract stage neither job has been submitted and no descriptor exists.

6.5 Stopping rule

Large-scale training is not authorized unless the low-cost mechanism tests pass. The programme stops or narrows if:

1. the structural score does not add transfer information beyond semantic similarity;

2. Q3 semantic decoys are frequently transferred;

3. structure labels are unstable across independent annotators or extraction methods;

4. a parent method matches the cost/capability frontier;
5. induction and verification cost eliminate any realistic break-even point; or
6. the training and inference systems require unrelated representations.

A null pilot is therefore useful: it prevents a large compute spend on a mechanism that has not survived its own easiest falsifiers.

The label-visible v1 proposal is not reusable as the confirmatory benchmark even if it later receives external annotations. Training authorization is evaluated only from the separately frozen v2 protocol, fresh label-blind items, externally provenanced annotations and adjudication, and the v2 held-out signal receipt. The batch-submission preflight binds all of those hashes to the exact Git subject and fails before invoking `sbatch` when any gate is absent or false.

6.6 RAKL v3 fresh-transfer evaluation

The downstream v3 experiment separates memory persistence from transfer validity. Both arms use the same underlying model, tools and frozen development experiences. A semantic-control arm retrieves and reuses candidate past experience using the strongest feasible content model; the witness arm applies the directional structural/boundary gate before the same promoted lessons or operators can influence routing. Every fresh-transfer task starts independently from the same frozen post-development state so one transfer case cannot teach the next.

Primary outcomes are target-task success/score, invalid-transfer or false-lesson reuse, repeated-failure rate, Q2 true accept, Q3 false accept and total resource use. Hostile near-misses are mandatory because an always-reuse policy can improve repeated-family tasks while increasing out-of-scope failures. A positive result requires material value beyond the strong content control under the same target verification rules; target scientific correctness cannot be inferred from the witness decision itself.

This experiment is distinct from the optional training-data redundancy hypothesis. Even if witness-gated experience reuse improves fresh inference, no claim about training-data efficiency follows without the separately authorized training experiment and full cost-to-capability accounting.

7 Engineering reporting contract

The empirical article must be interpretable as an AI-systems result, not only as a representation proposal. We therefore predeclare a reporting contract that couples capability, transfer validity and resource cost. The experimental unit is the task or evaluation item; repeated seeds are nested replicates. Principal comparisons freeze the base checkpoint, tokenizer, task set, held-out domains/families, context window, tool permissions, evaluator, budget schedule and seed schedule. Training-selection comparisons additionally freeze the downstream optimizer and training recipe; inference comparisons freeze the model weights.

7.1 Metrics that answer the engineering questions

For transfer discrimination, we report held-out ROC-AUC/PR-AUC, log loss or Brier score, calibration, Q2 true-accept and Q3 false-accept against the strongest semantic/skill control. The load-bearing quantity is incremental held-out value from witnessed structure, not raw accuracy of a structural classifier.

For training efficiency, let $T_m(q^*)$ be the minimum cumulative training-token exposure required by method m to reach a preregistered structural-OOD capability target q^*. The headline token reduction against parent p is

$$\Delta_T(q^*) = 1 - \frac{T_{\mathrm{RAKL}}(q^*)}{T_p(q^*)}.$$

We report the same target-based comparison for examples and, where measurable, training compute, together with a normalized area under the learning curve over a frozen token range. Structural-OOD is split into tasks with known structure but a new domain and the harder boundary in which whole structure families are held out; ordinary in-domain performance is retained as a safety check.

For inference efficiency, a *valid solve* requires task correctness plus the registered transfer and verification gates. Billable inference tokens sum provider-reported input and output tokens across retrieval planning, adaptation, criticism and verification calls. For stratum s,

$$\mathrm{TPVS}(s) = \frac{\sum_{i\in s} \text{billable inference tokens}_i}{\sum_{i\in s} \mathbf{1}[\text{valid solve}_i]}.$$

Zero valid solves therefore produce an infinite rather than missing cost-per-success value. We additionally report provider cost, tool calls, retrieval operations, median and p95 latency, and verification overhead.

The amortization claim uses cumulative total cost rather than online tokens alone:

$$\begin{aligned} C_{\text{total}}(n) = C_{\text{induction}} &+ C_{\text{training delta}} + C_{\text{storage}} \\ &+ C_{\text{retrieval}}(n) + C_{\text{adaptation}}(n) + C_{\text{reasoning}}(n) \\ &+ C_{\text{tools}}(n) + C_{\text{verification}}(n). \end{aligned}$$

The empirical break-even count is the smallest registered reuse count for which RAKL is no more costly than the comparator while the capability target and invalid-transfer ceiling remain satisfied. A workload with no break-even in the registered horizon is reported as such.

7.2 Main display logic

The empirical manuscript is planned around six claims rather than six data tables. Figure 1 explains the shared structural object and matched training/inference design without making a performance claim. Figure 2 tests whether witnessed structure predicts valid transfer beyond semantic/skill controls, with Q2 acceptance and Q3 false acceptance shown together. Figure 3 reports structural-OOD performance versus cumulative training tokens and tokens/compute to a frozen capability target. Figure 4 reports the frozen-model valid-success versus inference-cost frontier, including tokens per valid solve and Q2/Q3 safety. Figure 5 reports cumulative total cost and the empirical break-even region after induction, retrieval, adaptation and verification are charged. Figure 6 reports directionality, QoI, boundary, evidence and verification ablations as effect sizes rather than a decorative feature inventory.

All quantitative displays use editable vector text, final-size uncertainty definitions and task-level source data. They contain no arrows pointing to data points, no opaque text boxes over data and no point-by-point prose annotations. Panel labels remain outside the data region. Method names are carried by fixed-axis labels or a shared legend rather than text placed over marks. Main figures show the shortest evidence chain needed for the central claim; per-model, per-family, cache/reuse, token-decomposition, calibration and sensitivity analyses are reserved for Extended Data unless they become load bearing after outcomes are observed.

7.3 Representation ablations and shared-substrate criterion

The structural object is only justified if its fields are load bearing. We therefore ablate directionality, QoI, target boundaries, evidence identity, explicit non-preserved properties and post-transfer verification, and compare against replacing the witness with semantic similarity or a domain-specific skill label. Primary ablation outcomes are Q2 acceptance, Q3 false acceptance, valid task success, tokens per valid solve and break-even count.

The strongest shared-substrate claim requires the same structural identity/object type to be used in both data selection and inference reuse. If training and inference require unrelated representations, the evidence supports two narrower engineering mechanisms rather than one unifying substrate. If the representation improves validity but never reaches a realistic total-cost break-even, the paper must describe it as a reliability mechanism rather than an efficiency mechanism.

The corresponding shared-substrate claim is falsified when an executed preregistered necessary condition fails. Missing confirmatory annotation is instead missing evidence: it leaves that claim unlicensed without converting an unrun test into a negative result.

The full machine-readable metric and figure contract is frozen in `research/PAPER3_AI_ENGINEERING_METRICS_AND_FIGURES_20260810.md` before the model-level experiments are run.

Code, materials and AI-use disclosure

The public research artifact is maintained at `https://github.com/SzeChunYiu/RAKL`. The canonical manuscript source is the Paper III package under `publication/papers/`; frozen benchmark proposals, strong-control protocols, annotation schemas and machine-readable receipts remain in the research tree so that publication prose cannot silently replace missing evidence. Language models were used as research, coding and drafting tools; they are not authors. The constructed diagnostic labels are explicitly distinguished from genuinely external annotation, and no external judgement is represented as completed until a hash-bound response, adjudication and provenance audit exist. Author identity, affiliation and corresponding email are given on the title page. Funding, competing-interest, acknowledgement and licensing declarations remain author-supplied submission metadata and are not inferred by the repository.

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